Class 12 Mathematics

Chapter 8: Application of Integrals

Practice Set A: step-wise solutions

20 questions **53** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** CBSE SQP 2024-25 — 1 mark

A student observes an open-air honeybee nest on the branch of a tree, whose plane figure is parabolic in shape, given by $x^2 = 4y$. Then the area (in sq units) of the region bounded by the parabola $x^2 = 4y$ and the line $y = 4$ is:

(A) $\frac{32}{3}$ **(B)** $\frac{64}{3}$ **(C)** $\frac{128}{3}$ **(D)** $\frac{256}{3}$

Formula / idea: Area between a curve and the y-axis: $\int_c^d x\,dy$

1 Write x in terms of y (first quadrant)
$x^2 = 4y \Rightarrow x = 2\sqrt{y}$

2 The region is symmetric about the y-axis, so double the first-quadrant part
$A = 2\int_0^4 2\sqrt{y}\,dy$

3 Integrate and put the limits ✓ 1
$A = 4 \cdot \frac{2}{3}\left[y^{3/2}\right]_0^4 = 4 \cdot \frac{2}{3} \cdot 8 = \frac{64}{3}$

4 So the area is $\frac{64}{3}$ sq units, option (B).

Answer **(B)** $\frac{64}{3}$

Watch out: Forgetting to double: the first-quadrant part alone is $\frac{32}{3}$, which is option (A).

2 MCQ Board-pattern 1 mark

The area of the region in the first quadrant enclosed by the circle $x^2 + y^2 = 36$ and the coordinate axes is:

(A) 9π **sq units** **(B)** 18π sq units

(C) 36π sq units **(D)** 6π sq units

Formula / idea: $\displaystyle\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$

1 Radius $r = 6$; in the first quadrant the curve is
$y = \sqrt{36 - x^2}, \quad 0 \le x \le 6$

2 Write the area
$A = \displaystyle\int_0^6 \sqrt{36 - x^2}\,dx = \left[\frac{x}{2}\sqrt{36 - x^2} + 18\sin^{-1}\frac{x}{6}\right]_0^6$

3 Put the limits ✓ 1
$A = 0 + 18 \cdot \frac{\pi}{2} - 0 = 9\pi$

4 So the area is 9π sq units, option (A).

Answer **(A)** 9π **sq units**

Watch out: Giving the area of the whole circle, $\pi r^2 = 36\pi$, when only one quadrant is asked.

3 MCQ Board-pattern 1 mark

The area enclosed by the ellipse $x^2 + 4y^2 = 16$ is:

(A) 16π sq units **(B)** 64π sq units

(C) 8π **sq units** **(D)** 4π sq units

Formula / idea: Area enclosed by $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ is πab.

1 Divide by 16 to get the standard form
$\dfrac{x^2}{16} + \dfrac{y^2}{4} = 1 \Rightarrow a = 4,\ b = 2$

2 Use area $= \pi ab$ ✓ 1
$A = \pi \times 4 \times 2 = 8\pi$

3 So the area is 8π sq units, option (C).

Answer **(C)** 8π **sq units**

Watch out: Using a^2 and b^2 instead of a and b: $\pi \times 16 \times 4 = 64\pi$.

4 MCQ Board-pattern 1 mark

The area of the region bounded by the curve $y = \sin x$, the x-axis and the lines $x = 0$ and $x = \pi$ is:

(A) 1 sq unit **(B)** **2 sq units** **(C)** 0 **(D)** π sq units

1 $\sin x \geq 0$ on $[0, \pi]$, so the area is the integral itself

$$A = \int_0^{\pi} \sin x \, dx$$

2 Integrate and put the limits ✓ 1

$$A = [-\cos x]_0^{\pi} = -\cos \pi + \cos 0 = 1 + 1 = 2$$

3 So the area is 2 sq units, option (B).

Answer **(B) 2 sq units**

Watch out: Writing $-\cos \pi$ as -1; since $\cos \pi = -1$, $-\cos \pi = +1$.

5 MCQ Board-pattern 1 mark

The area of the region bounded by the line $x + 2y = 4$ and the coordinate axes is:

(A) 8 sq units **(B)** 2 sq units

(C) 16 sq units **(D)** **4 sq units**

1 Write y in terms of x; the line meets the x-axis at $x = 4$

$$y = \frac{4 - x}{2}$$

2 Area under the line from $x = 0$ to $x = 4$

$$A = \int_0^4 \frac{4 - x}{2} \, dx = \frac{1}{2}\left[4x - \frac{x^2}{2}\right]_0^4$$

3 Put the limits ✓ 1

$$A = \frac{1}{2}(16 - 8) = 4$$

4 So the area is 4 sq units, option (D).

Answer **(D) 4 sq units**

Watch out: Multiplying the intercepts, $4 \times 2 = 8$, and forgetting the $\frac{1}{2}$.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): The area of the region bounded by the curve $y = x^3$, the x-axis and the lines $x = -1$ and $x = 1$ is 0. **Reason (R):** $\int_{-1}^{1} x^3\,dx = 0$, as x^3 is an odd function.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) **Assertion (A) is false, but Reason (R) is true.**

Formula / idea: Area is always positive: take the modulus of the part below the x-axis.

1 For A: $x^3 < 0$ on $(-1, 0)$, so split the area

$$\text{Area} = \left|\int_{-1}^{0} x^3\,dx\right| + \int_{0}^{1} x^3\,dx = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

2 The area is $\frac{1}{2}$, not 0, so A is false

3 For R: evaluate the integral

$$\int_{-1}^{1} x^3\,dx = \left[\frac{x^4}{4}\right]_{-1}^{1} = \frac{1}{4} - \frac{1}{4} = 0$$

Integral of an odd function over $[-a, a]$ is 0.

4 So R is true; R cannot explain A, since A is false

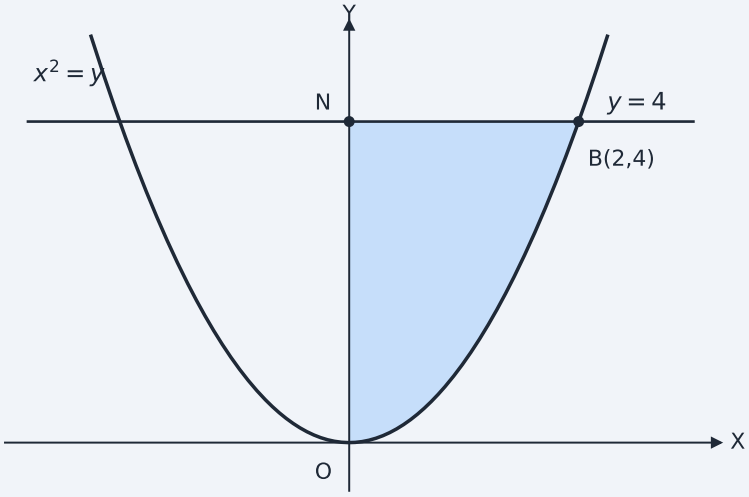
5 A is false and R is true, so the option is (D) ✓ 1

Answer **(D) A is false, but R is true**

Watch out: Taking the definite integral as the area; the two equal parts cancel in the integral.

7 **Short answer** CBSE SQP 2025-26 2 marks

Find the area of the shaded region in the figure given below. The region lies in the first quadrant and is bounded by the parabola $x^2 = y$, the y-axis and the line $y = 4$.



Formula / idea: Area between a curve and the y-axis: $A = \int_c^d x\,dy$

1 In the first quadrant, $x = \sqrt{y}$; take horizontal strips from $y = 0$ to $y = 4$

$$A = \int_0^4 x\,dy = \int_0^4 \sqrt{y}\,dy$$

✓ 1

2 Integrate and put the limits

$$A = \frac{2}{3}\left[y^{3/2}\right]_0^4 = \frac{2}{3}(8 - 0)$$

3 Required area

$$A = \frac{16}{3} \text{ sq units}$$

✓ 1

Answer $\frac{16}{3}$ **sq units**

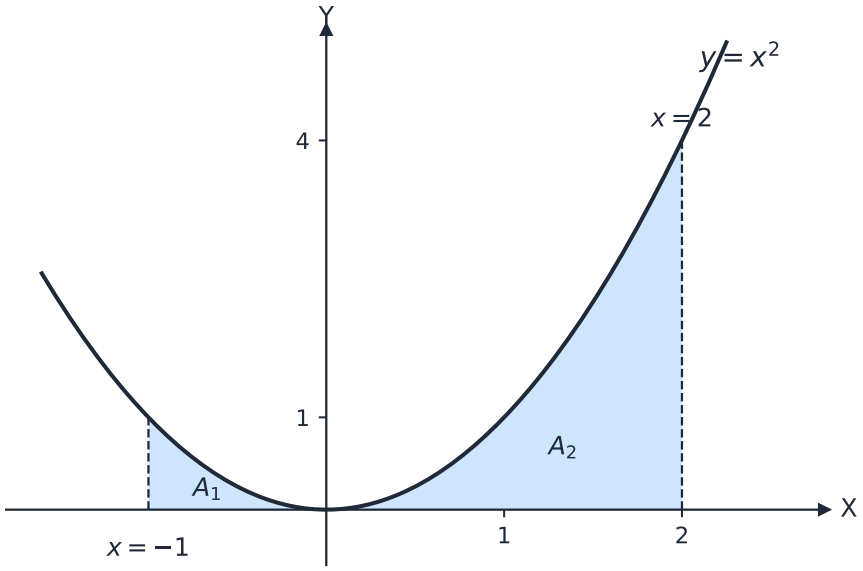
Watch out: Finding $\int_0^2 x^2\,dx = \frac{8}{3}$: that is the area below the curve, not the shaded part.

Level 2

18 marks

8 **Short answer** PYQ-style 3 marks

Sketch the graph of $y = x^2$ and, using integration, find the area of the region bounded by the curve $y = x^2$, the x-axis and the lines $x = -1$ and $x = 2$.



1	Sketch the parabola $y = x^2$ and shade the region from $x = -1$ to $x = 2$	✓ ½
2	$y = x^2 \geq 0$ for all x, so the whole region is above the x-axis $A = \int_{-1}^{2} x^2 \, dx$	✓ 1
3	Integrate and put the limits $A = \left[\frac{x^3}{3}\right]_{-1}^{2} = \frac{8}{3} - \left(-\frac{1}{3}\right)$	✓ 1
4	Required area $A = \frac{9}{3} = 3$ sq units	✓ ½

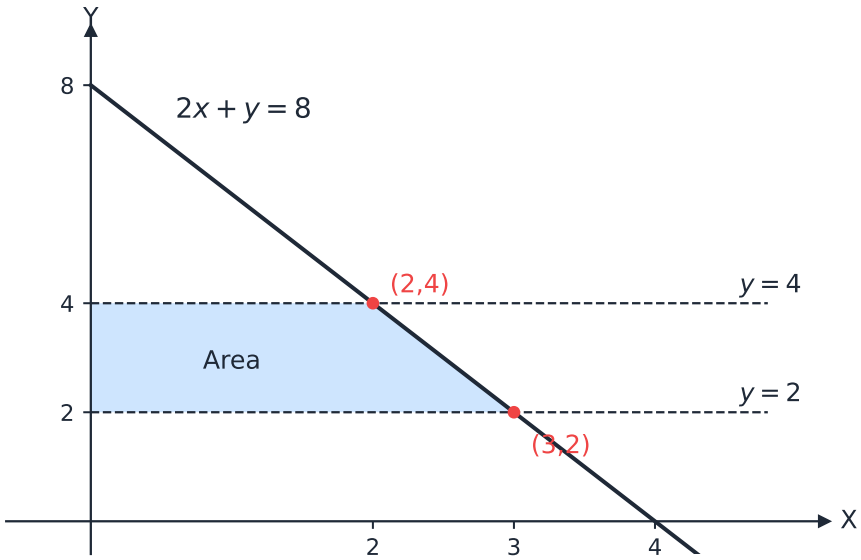
Answer 3 **sq units**

Watch out: Subtracting $\frac{1}{3}$ instead of $-\frac{1}{3}$ at the lower limit, which gives $\frac{7}{3}$.

9 **Short answer** PYQ-style 3 marks

Sketch the region bounded by the lines $2x + y = 8$, $y = 2$, $y = 4$ and the y-axis. Hence, obtain its area using integration.

Formula / idea: Area between a curve and the y-axis from $y = c$ to $y = d$ is $\int_c^d x\,dy$.



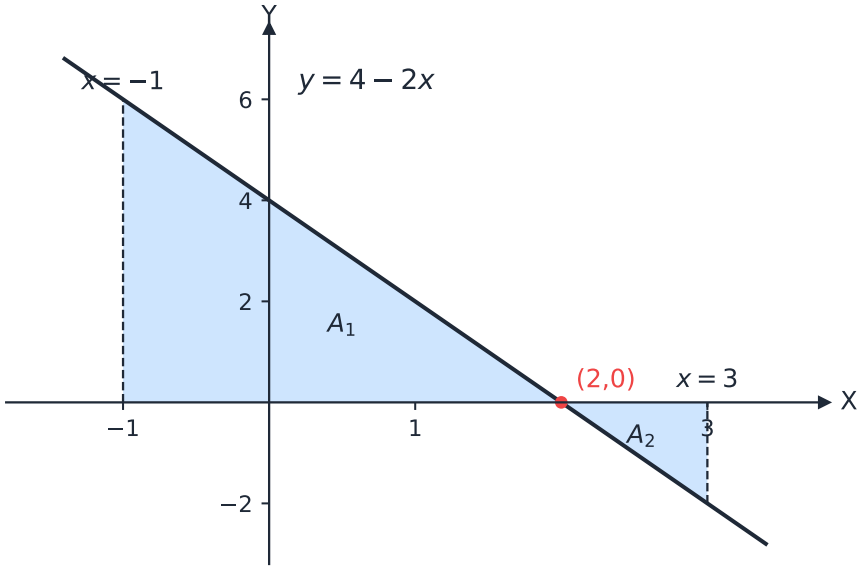
1	Sketch: the line meets $y = 2$ at $(3, 2)$ and $y = 4$ at $(2, 4)$	✓ 1
2	Write x in terms of y $2x + y = 8 \Rightarrow x = \dfrac{8 - y}{2}$	
3	Take horizontal strips from $y = 2$ to $y = 4$ $A = \int_2^4 x\,dy = \dfrac{1}{2}\int_2^4 (8 - y)\,dy$	✓ 1
4	Integrate and put the limits $A = \dfrac{1}{2}\left[8y - \dfrac{y^2}{2}\right]_2^4 = \dfrac{1}{2}\left[(32 - 8) - (16 - 2)\right]$	✓ ½
5	Required area $A = \dfrac{1}{2} \times 10 = 5$ sq units	✓ ½

Answer 5 **sq units**

Watch out: Using 2 and 4 as x-limits in $\int y\,dx$; they are y-values, so integrate $x\,dy$.

10 **Short answer** PYQ-style 3 marks

Using integration, find the area of the region bounded by the line $y = 4 - 2x$, the x-axis and the ordinates $x = -1$ and $x = 3$.



1	Find where the line meets the x-axis (put $y = 0$) $4 - 2x = 0 \Rightarrow x = 2$	✓ ½
2	Line is above the x-axis on $[-1, 2]$ and below it on $[2, 3]$ $A = \int_{-1}^{2} (4 - 2x)\,dx + \left\lvert \int_{2}^{3} (4 - 2x)\,dx \right\rvert$	✓ 1
3	First integral $\left[4x - x^2\right]_{-1}^{2} = (8 - 4) - (-4 - 1) = 9$	✓ ½
4	Second integral $\left[4x - x^2\right]_{2}^{3} = (12 - 9) - (8 - 4) = -1$	✓ ½
5	Required area $A = 9 + \lvert -1 \rvert = 10$ sq units	✓ ½

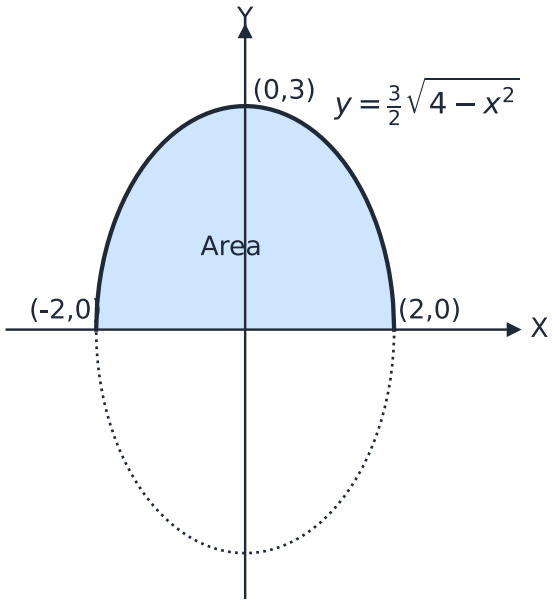
Answer 10 **sq units**

Watch out: Integrating straight from -1 to 3 gives $9 - 1 = 8$; the part below the x-axis must be added as $\lvert -1 \rvert$.

11 **Short answer** Board-pattern 2 marks

Using integration, find the area of the region bounded by the curve $y = \frac{3}{2}\sqrt{4 - x^2}$ and the x-axis.

Formula / idea: $\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$



1	Square both sides: the curve is the upper half of an ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1, \quad y \geq 0$	
2	It meets the x-axis at $x = \pm 2$; use symmetry about the y-axis $A = \int_{-2}^{2} \frac{3}{2}\sqrt{4 - x^2}\,dx = 3\int_{0}^{2} \sqrt{4 - x^2}\,dx$	✓ 1
3	Integrate and put the limits $A = 3\left[\frac{x}{2}\sqrt{4 - x^2} + 2\sin^{-1}\frac{x}{2}\right]_0^2 = 3\left[0 + 2 \cdot \frac{\pi}{2}\right]$	✓ ½
4	Required area $A = 3\pi$ sq units	✓ ½

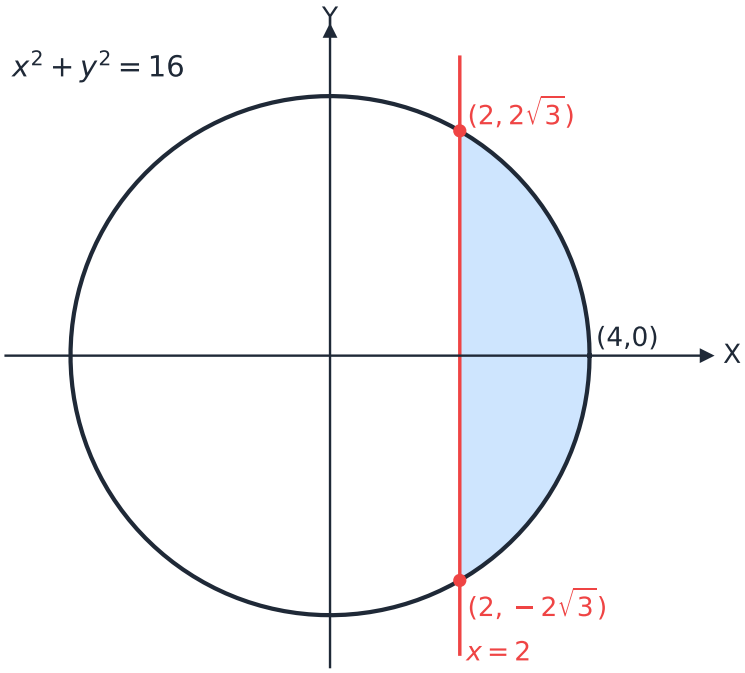
Answer 3π **sq units**

Watch out: Taking the whole ellipse, 6π; the curve $y = \frac{3}{2}\sqrt{4 - x^2}$ gives only the upper half.

12 **Short answer** PYQ-style 3 marks

Using integration, find the area of the minor segment of the circle $x^2 + y^2 = 16$ cut off by the line $x = 2$.

Formula / idea: $\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$



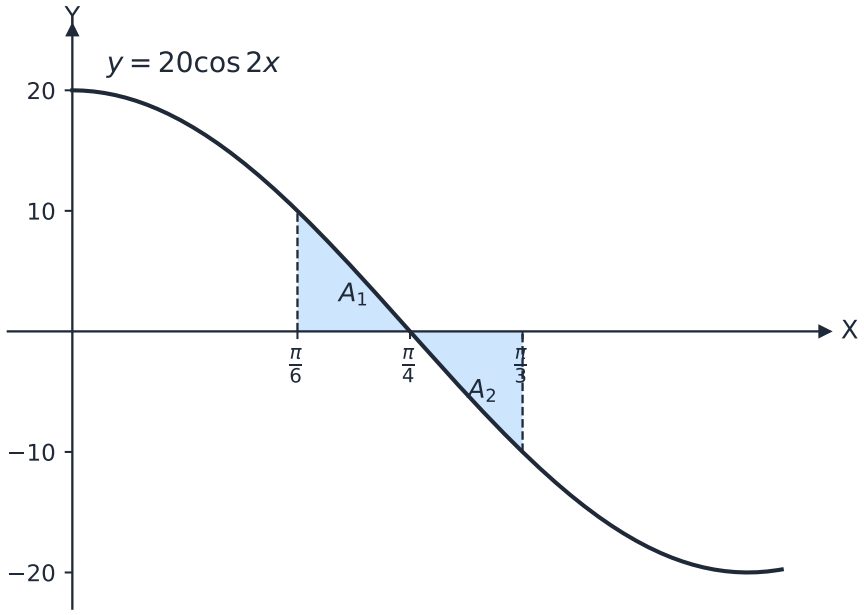
1	The line $x = 2$ meets the circle at $(2, \pm 2\sqrt{3})$; the minor segment runs from $x = 2$ to $x = 4$	✓ ½
2	The segment is symmetric about the x-axis $A = 2\int_2^4 \sqrt{16 - x^2}\,dx$	✓ 1
3	Use the formula for $\int \sqrt{a^2 - x^2}\,dx$ $A = 2\left[\frac{x}{2}\sqrt{16 - x^2} + 8\sin^{-1}\frac{x}{4}\right]_2^4$	✓ ½
4	Put the limits $A = 2\left[\left(0 + 8\cdot\frac{\pi}{2}\right) - \left(2\sqrt{3} + 8\cdot\frac{\pi}{6}\right)\right]$	✓ ½
5	Simplify $A = 2\left[4\pi - 2\sqrt{3} - \frac{4\pi}{3}\right] = 2\left[\frac{8\pi}{3} - 2\sqrt{3}\right]$	
6	Required area $A = \left(\frac{16\pi}{3} - 4\sqrt{3}\right)$ sq units	✓ ½

Answer $\left(\frac{16\pi}{3} - 4\sqrt{3}\right)$ **sq units**

Watch out: Forgetting the factor 2: the integral alone gives only the half above the x-axis.

13 **Short answer** CBSE SQP 2024-25 2 marks

Draw the rough sketch of the curve $y = 20\cos 2x$, where $\frac{\pi}{6} \le x \le \frac{\pi}{3}$. Using integration, find the area of the region bounded by the curve $y = 20\cos 2x$ from the ordinates $x = \frac{\pi}{6}$ to $x = \frac{\pi}{3}$ and the x-axis.



1 Sketch: the curve cuts the x-axis where $\cos 2x = 0$

$2x = \frac{\pi}{2} \Rightarrow x = \frac{\pi}{4}$ ✓ ½

2 Curve is above the x-axis on $\left[\frac{\pi}{6}, \frac{\pi}{4}\right]$ and below on $\left[\frac{\pi}{4}, \frac{\pi}{3}\right]$

$A = 20\int_{\pi/6}^{\pi/4} \cos 2x\, dx + \left|20\int_{\pi/4}^{\pi/3} \cos 2x\, dx\right|$ ✓ ½

3 Integrate

$A = 10\,[\sin 2x]_{\pi/6}^{\pi/4} + \left|10\,[\sin 2x]_{\pi/4}^{\pi/3}\right|$

4 Put the limits

$A = 10\left(1 - \frac{\sqrt{3}}{2}\right) + \left|10\left(\frac{\sqrt{3}}{2} - 1\right)\right|$ ✓ ½

5 Required area

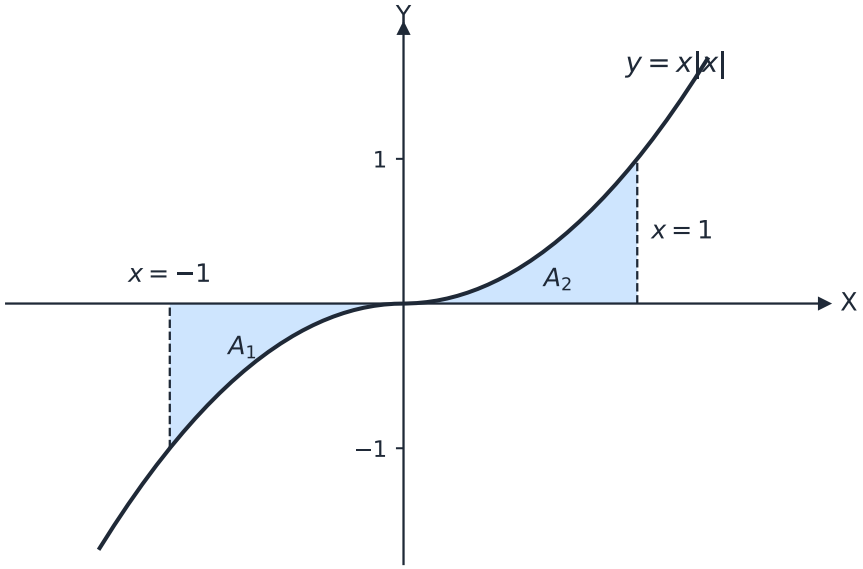
$A = (10 - 5\sqrt{3}) + (10 - 5\sqrt{3}) = (20 - 10\sqrt{3})$ sq units ✓ ½

Answer $(20 - 10\sqrt{3})$ **sq units**

Watch out: Integrating straight from $\frac{\pi}{6}$ to $\frac{\pi}{3}$ gives 0, because the two parts cancel.

14 **Short answer** PYQ-style 2 marks

Using integration, find the area of the region bounded by the curve $y = x|x|$, the x-axis and the ordinates $x = -1$ and $x = 1$.



1 Remove the modulus

$$y = x|x| = \begin{cases} -x^2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

✓ ½

2 The curve is below the x-axis for $x < 0$, so take the modulus there

$$A = \left| \int_{-1}^{0} (-x^2)\, dx \right| + \int_{0}^{1} x^2\, dx$$

✓ ½

3 Integrate and put the limits

$$A = \left| \left[-\frac{x^3}{3} \right]_{-1}^{0} \right| + \left[\frac{x^3}{3} \right]_{0}^{1} = \left| -\frac{1}{3} \right| + \frac{1}{3}$$

✓ ½

4 Required area

$$A = \frac{2}{3} \text{ sq units}$$

✓ ½

Answer $\frac{2}{3}$ **sq units**

Watch out: Writing the area as $\int_{-1}^{1} x|x|\, dx = 0$; $x|x|$ is odd, so the two parts cancel.

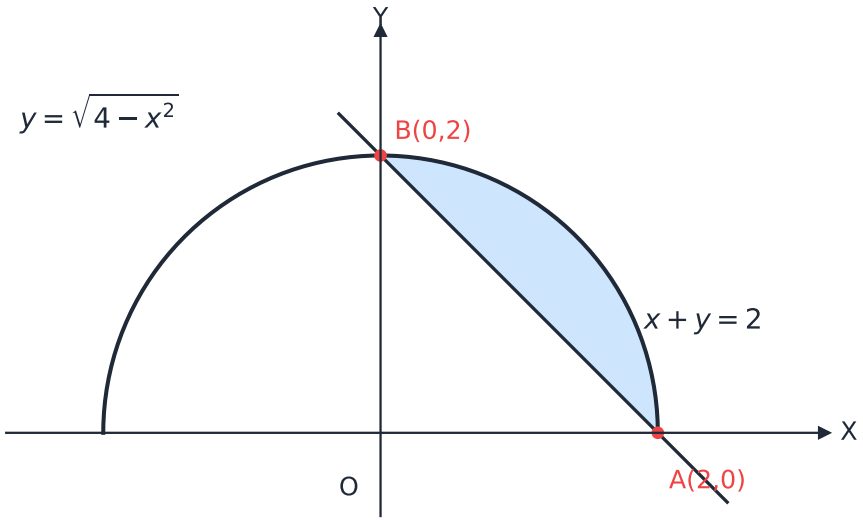
Level 3

27 marks

15 **Long answer** CBSE SQP 2026-27 5 marks

Using integration, find the area of the region $\{(x, y) : 0 \le y \le \sqrt{4 - x^2},\ x \ge 2 - y\}$.

Formula / idea: $\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$



1	$y = \sqrt{4 - x^2}$ is the upper semicircle of $x^2 + y^2 = 4$ (centre O, radius 2)	
2	$x \ge 2 - y$ means $x + y \ge 2$: the side of the line away from O	
3	The line $x + y = 2$ meets the circle at $A(2, 0)$ and $B(0, 2)$; sketch and shade the region	✓ 1
4	The circle is on top and the line below, for $0 \le x \le 2$ $A = \int_0^2 \left[\sqrt{4 - x^2} - (2 - x)\right] dx$	✓ 1
5	Integrate $A = \left[\frac{x}{2}\sqrt{4 - x^2} + 2\sin^{-1}\frac{x}{2} - 2x + \frac{x^2}{2}\right]_0^2$	✓ 1
6	Put the limits $A = \left(0 + 2 \cdot \frac{\pi}{2} - 4 + 2\right) - 0$	✓ 1
7	Required area $A = (\pi - 2)$ sq units	✓ 1

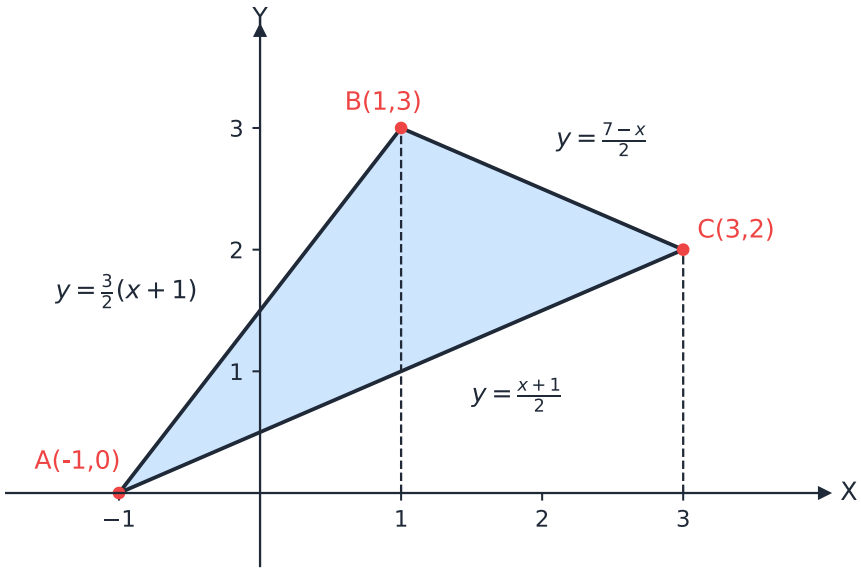
Answer $(\pi - 2)$ **sq units**

Watch out: Shading the triangle OAB instead: $x \ge 2 - y$ is the side of the line away from the origin.

16 **Long answer** PYQ-style 5 marks

Using integration, find the area of the triangle whose vertices are $A(-1,0)$, $B(1,3)$ and $C(3,2)$.

Formula / idea: Line through two points: $y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)$.



1	Plot the three vertices and draw the triangle	✓ ½
2	Equation of AB (two-point form) $y - 0 = \dfrac{3-0}{1+1}(x+1) \Rightarrow y = \dfrac{3}{2}(x+1)$	✓ ½
3	Equation of BC $y - 3 = \dfrac{2-3}{3-1}(x-1) \Rightarrow y = \dfrac{7-x}{2}$	✓ ½
4	Equation of AC $y - 0 = \dfrac{2-0}{3+1}(x+1) \Rightarrow y = \dfrac{x+1}{2}$	✓ ½
5	Area = area under AB + area under BC − area under AC $A = \displaystyle\int_{-1}^{1} \frac{3}{2}(x+1)\,dx + \int_{1}^{3} \frac{7-x}{2}\,dx - \int_{-1}^{3} \frac{x+1}{2}\,dx$	✓ 1
6	First integral $\dfrac{3}{2}\left[\dfrac{(x+1)^2}{2}\right]_{-1}^{1} = \dfrac{3}{2} \times 2 = 3$	✓ ½
7	Second integral $\dfrac{1}{2}\left[7x - \dfrac{x^2}{2}\right]_{1}^{3} = \dfrac{1}{2}\left[\dfrac{33}{2} - \dfrac{13}{2}\right] = 5$	✓ ½
8	Third integral $\dfrac{1}{2}\left[\dfrac{(x+1)^2}{2}\right]_{-1}^{3} = \dfrac{1}{2} \times 8 = 4$	✓ ½

9	Required area $A = 3 + 5 - 4 = 4$ sq units	✓ ½

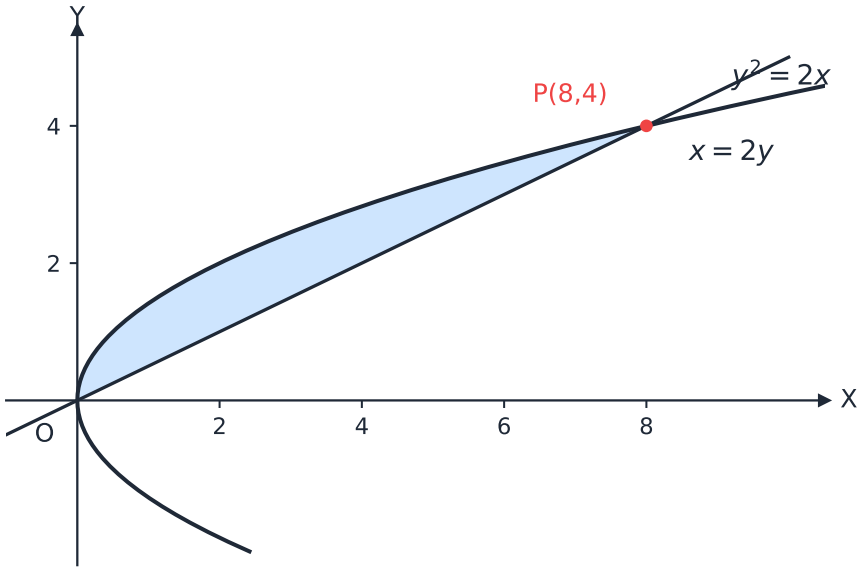
Answer 4 **sq units**

Watch out: Adding the area under AC instead of subtracting it; AC is the lower side of the triangle.

17 **Long answer** CBSE SQP 2026-27 5 marks

Using integration, find the area of the region bounded by the curve $y^2 = 2x$ and the line $x - 2y = 0$.

Formula / idea: Area between two curves $= \int_a^b (y_{\text{upper}} - y_{\text{lower}})\, dx$.



1	Put $x = 2y$ in $y^2 = 2x$ $y^2 = 4y \Rightarrow y = 0$ or $y = 4$	
2	So the curves meet at $O(0,0)$ and $P(8,4)$	✓ 1
3	Sketch: between O and P the parabola $y = \sqrt{2x}$ lies above the line $y = \dfrac{x}{2}$	✓ 1
4	Write the area $A = \int_0^8 \left(\sqrt{2x} - \dfrac{x}{2}\right) dx$	✓ 1
5	Integrate $A = \left[\sqrt{2} \cdot \dfrac{2}{3} x^{3/2} - \dfrac{x^2}{4}\right]_0^8$	✓ 1
6	Put the limits, using $8^{3/2} = 16\sqrt{2}$ $A = \dfrac{2\sqrt{2}}{3} \cdot 16\sqrt{2} - 16 = \dfrac{64}{3} - 16$	

7 Required area

$$A = \frac{16}{3} \text{ sq units}$$

✓ 1

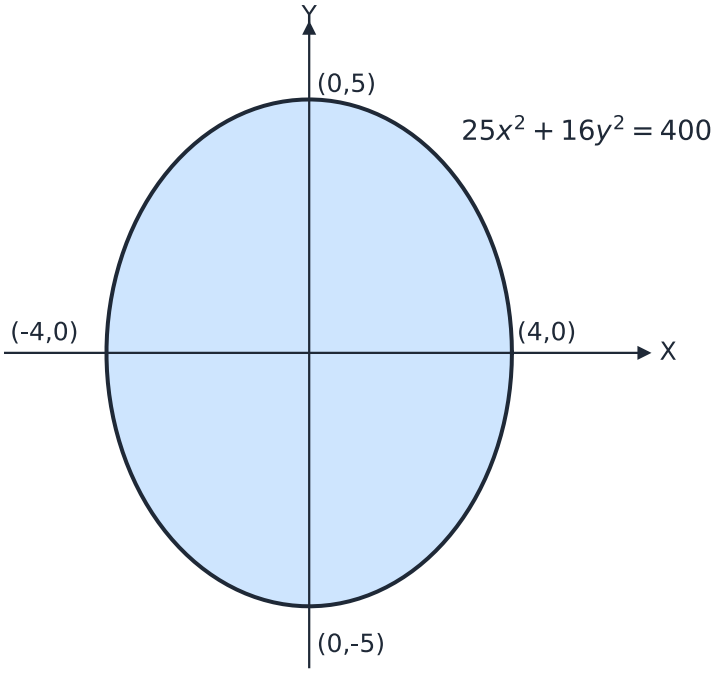
Answer $\frac{16}{3}$ **sq units**

Watch out: Taking the line as the upper curve gives $-\frac{16}{3}$; check which curve is on top first.

18 **Long answer** CBSE SQP 2026-27 5 marks

Using integration, find the area of the region enclosed within $25x^2 + 16y^2 = 400$.

Formula / idea: $\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$



1 Divide by 400 to get the standard form

$$\frac{x^2}{16} + \frac{y^2}{25} = 1, \quad a = 4,\ b = 5$$

✓ ½

2 Sketch the ellipse through $(\pm 4, 0)$ and $(0, \pm 5)$ ✓ 1

3 In the first quadrant

$$y = \frac{5}{4}\sqrt{16 - x^2}$$

4 The ellipse is symmetric about both axes

$$A = 4\int_0^4 \frac{5}{4}\sqrt{16 - x^2}\, dx = 5\int_0^4 \sqrt{16 - x^2}\, dx$$

✓ 1

5 Integrate

$$A = 5\left[\frac{x}{2}\sqrt{16 - x^2} + 8\sin^{-1}\frac{x}{4}\right]_0^4$$

✓ 1

6 Put the limits
$A = 5\left[0 + 8 \cdot \frac{\pi}{2} - 0\right] = 5 \times 4\pi$ ✓ ½

7 Required area
$A = 20\pi$ sq units ✓ 1

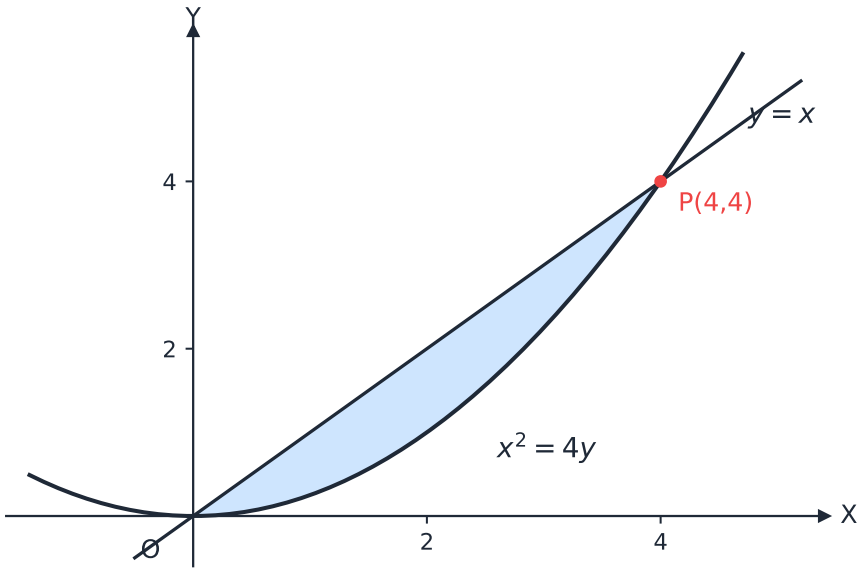
Answer 20π **sq units**

Watch out: Writing $y = \frac{4}{5}\sqrt{25 - x^2}$; here the larger denominator, 25, is under y^2.

19 **Short answer** CBSE SQP 2025-26 3 marks

Using integration, find the area of the region $\{(x, y) : x^2 - 4y \leq 0,\ y - x \leq 0\}$.

Formula / idea: Area between two curves $= \int_a^b (y_{\text{upper}} - y_{\text{lower}})\, dx$.



1 $x^2 - 4y \leq 0$ means $y \geq \frac{x^2}{4}$: on or above the parabola $x^2 = 4y$

2 $y - x \leq 0$ means $y \leq x$: on or below the line $y = x$

3 Put $y = x$ in $x^2 = 4y$ to get the corners
$x^2 = 4x \Rightarrow x = 0,\ 4; \quad O(0, 0),\ P(4, 4)$ ✓ 1

4 Line on top, parabola below, for $0 \leq x \leq 4$
$A = \int_0^4 x\, dx - \int_0^4 \frac{x^2}{4}\, dx$ ✓ 1

5 Integrate and put the limits
$A = \left[\frac{x^2}{2}\right]_0^4 - \frac{1}{12}\left[x^3\right]_0^4 = 8 - \frac{16}{3}$ ✓ ½

6 Required area

$$A = \frac{8}{3} \text{ sq units}$$

✓ ½

Answer $\frac{8}{3}$ **sq units**

Watch out: Reading $y - x \leq 0$ as the region above the line; it means $y \leq x$, below the line.

20 **Case study** Board-pattern 4 marks

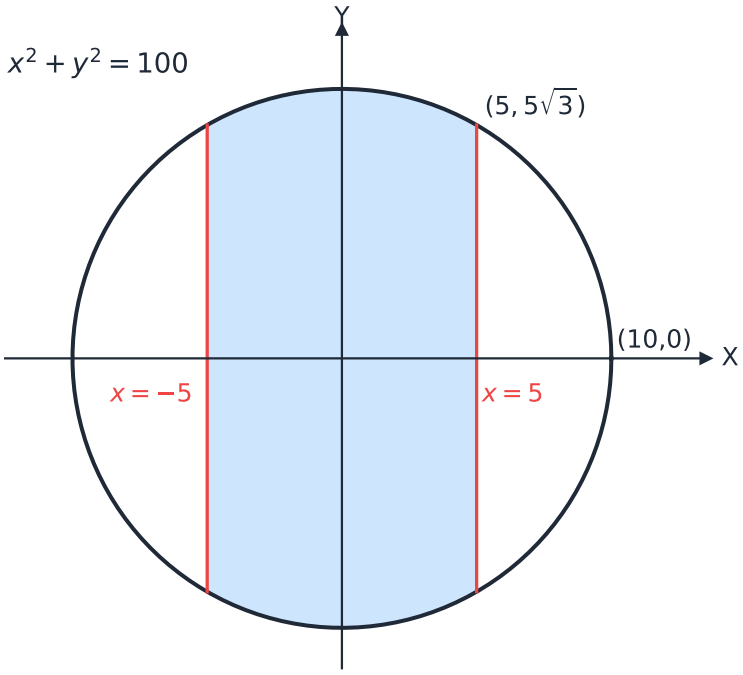
Case Study: A circular garden in a housing society has its boundary given by $x^2 + y^2 = 100$, with its centre at the origin (units in metres). A flower bed is to be made in the part of the garden between two straight paths along the lines $x = -5$ and $x = 5$. Based on this information, answer the following questions using integration.

(i) Find the total area of the garden. 1 mark

(ii) Find the area of the part of the garden lying in the first quadrant. 1 mark

(iii) Find the area of the flower bed, that is, the part of the garden between the lines $x = -5$ and $x = 5$. 2 marks

Formula / idea: $\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$



(i) 1 mark

1 Radius $= 10$; the circle is symmetric about both axes

$$A = 4\int_0^{10} \sqrt{100 - x^2}\, dx = 4\left[\frac{x}{2}\sqrt{100 - x^2} + 50\sin^{-1}\frac{x}{10}\right]_0^{10}$$

✓ ½

2 Put the limits

$$A = 4 \times 50 \cdot \frac{\pi}{2} = 100\pi \text{ m}^2$$

✓ ½

Answer (i) $100\pi \text{ m}^2$

(ii)

1 mark

1 Take the first-quadrant part only

$$A = \int_0^{10} \sqrt{100 - x^2}\, dx = \left[\frac{x}{2}\sqrt{100 - x^2} + 50 \sin^{-1} \frac{x}{10}\right]_0^{10}$$

✓ ½

2 Put the limits

$$A = 50 \cdot \frac{\pi}{2} = 25\pi \text{ m}^2$$

✓ ½

Answer (ii) $25\pi \text{ m}^2$

(iii)

2 marks

1 The bed is symmetric about both axes: four equal parts

$$A = 4 \int_0^{5} \sqrt{100 - x^2}\, dx$$

✓ ½

2 Integrate

$$A = 4 \left[\frac{x}{2}\sqrt{100 - x^2} + 50 \sin^{-1} \frac{x}{10}\right]_0^{5}$$

✓ ½

3 Put the limits, using $\sqrt{75} = 5\sqrt{3}$ and $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$

$$A = 4 \left[\frac{5}{2} \cdot 5\sqrt{3} + 50 \cdot \frac{\pi}{6}\right] = 4 \left[\frac{25\sqrt{3}}{2} + \frac{25\pi}{3}\right]$$

✓ ½

4 Required area

$$A = \left(50\sqrt{3} + \frac{100\pi}{3}\right) \text{ m}^2$$

✓ ½

Answer (iii) $\left(50\sqrt{3} + \frac{100\pi}{3}\right) \text{ m}^2$

Watch out: In (iii), finding only the first-quadrant strip; the flower bed has four equal parts.